

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electrical Engineering Summer Term Examination Academic Year: 2024-25 (Semester – I)	
Class: TY B.Tech	Subject: Microprocessor and Microcontroller	Subject Code: PCC-EE 303
Date: 19.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed
- Non Programmable calculator is allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.	Attempt The Following Questions	Marks	CO	BT
Q.1	What are types of instruction set in 8085. Explain any two instructions of each type.	10	CO1	BT1
Q.2	Explain the functional block diagram of 8085 microprocessor in detail.	10	CO1	BT1
Q.3	a. Draw timing diagram for instruction IN 80 H. Opcode for given instruction is DB H.	8	CO1	BT3
	b. Explain in detail timer modes in 8051.	7	CO2	BT1
Q.4	a. Interface two chips of 4K RAM and one chip of 2K ROM with microprocessor. Starting address of RAM is 8000H.	8	CO2	BT3
	b. Subtract the 16 bit number in memory location 3002H and 3003H from the 16 bit number in 3000H and 3001H. The most significant eight bits of two numbers are in memory location 3001H and 3003H. Store the result in memory locations 3004H and 3005H with most significant byte in memory location 3005H.	7	CO3	BT3
Q.5	a. Write an assembly language program to add two numbers (decimal) 38 and 64, then subtract decimal number 3 from the sum. The final answer is to be stored in memory location 2100 H.	5	CO3	BT3
	b. Write assembly language program to subtract two 8-bit numbers and Display Borrow.	5	CO3	BT3
Q.6	a. What are interrupt sources in 8051 microcontroller?	7	CO4	BT2
	b. Explain in detail architecture of programmable peripheral interface (PPI).	8	CO4	BT2
Q.7	a. Describe working of Programmable display controller.	10	CO4	BT3
Q.8	a. Explain in detail interfacing of 8051 with ADC.	10	CO4	BT1
	b. Explain in detail following instructions in 8051 JUMP, CALL	5	CO5	BT2

== ALL THE BEST ==